

Unit 08: Entrepreneurship in Digital Age

After completing this unit you will be able to:

- Understand what prototyping is and why it is important.
- Create, test and iterate a prototype for a business idea.

8.1 Introduction to Product Development

Starting a business (a startup) is easy, especially for young professionals who are yet to start their career. However, entrepreneurship requires skills, qualities and traits to take the risks of launching a business and taking off with success. The field of product development is dynamic and offers many opportunities to people who are passionate about creating impactful and user-centred solutions in any industry (technology, healthcare, entertainment, etc.).

8.1.1 Definition of a Product or Service

A product or service is the fundamental outcome of a process designed to meet a specific need or want of consumers. Products and services are tangible or intangible offerings that are created, developed and delivered to address a particular problem, enhance convenience, provide entertainment, or fulfil any other customer requirement.

- **Products:** Typically physical items that can be seen, touched or held. They range from everyday goods to industrial machinery.
 - *Example:* smartphones, clothing, automobiles, industrial machinery and consumer electronics.
- **Services:** Intangible and often involve a specific action or task performed by individuals or organizations.
 - *Example:* healthcare services, consulting, education and entertainment streaming.
- **Phases associated with products/services (Fig 8.1):** Products go through Design, Manufacturing and After-Sales Services. Services are linked with fields such as Educational Technology, Medical devices and Enterprise Business Technology.

8.1.2 Importance of Successful Product Development

Successful product development is a cornerstone of business sustainability and growth. It is not just about bringing a new item to the market; it is about doing so effectively and efficiently.

- **Competitive Advantage:** A well-developed product can set a company apart from competitors by offering unique features, better quality or a more appealing price.
 - *Example:* A phone brand with a longer battery life becomes the preferred choice of customers.
- **Revenue Generation:** New products and services open up revenue streams, expanding the customer base and increasing profits.
- **Customer Satisfaction:** By addressing the specific needs and wants of consumers, successful products enhance satisfaction and loyalty.
- **Innovation and Adaptation:** Product development keeps a company at the forefront of its industry and helps it adapt to changing market dynamics.
- **Market Expansion:** Developing new products lets a company explore new markets and demographics, diversifying its business.

8.2 Understanding Prototypes

8.2.1 What is a Prototype?

A prototype is a tangible or digital representation of a product (or a part of it), created during the design and development phase. It is a preliminary model that allows designers, engineers and stakeholders to visualize, test and refine the concept **before committing to full-scale production**. Prototypes vary in fidelity, complexity and purpose, but they all share the common goal of providing valuable insights and feedback to improve the final product.

Types of Prototypes

- **1. Low-Fidelity Prototypes (minimum visual details):**
 - **Sketches and Wireframes:** simple, hand-drawn or digital representations that focus on the basic layout, structure and flow of a product.
 - Quick and inexpensive to create; ideal for early-stage ideation and concept validation.
 - *Example:* A hand-drawn sketch of the screens of a food delivery app.
- **2. Medium-Fidelity Prototypes:**
 - **Paper Prototypes:** physical, interactive models made with paper, cardboard or other easily manipulated materials.
 - More detailed than sketches; suitable for user testing and gathering initial feedback.
 - *Example:* Paper cut-outs of app screens that a tester 'taps' while a team member swaps the pages.
- **3. High-Fidelity Prototypes:**
 - **Digital Prototypes:** interactive, computer-generated simulations of the final product.
 - They closely resemble the user interface and functionality of the actual product, so they are valuable for user testing, usability studies and investor or stakeholder presentations.
 - *Example:* A clickable Figma prototype that looks and behaves like the real app.
- **4. Functional Prototypes:**
 - **Proof-of-Concept Prototypes:** demonstrate the core functionality and technical feasibility of a product but may lack the full aesthetic design.
 - *Example:* A basic circuit that shows a sensor can detect pressure, before any casing or design is made.
- **5. Working Prototypes:**
 - **Alpha Prototypes:** advanced, functional versions with most features in place; used for internal testing and refinement.
 - **Beta Prototypes:** near-final versions made available to a select group of users or testers to identify remaining issues or bugs before the official release.
 - *Example:* A game released to a few hundred players as a beta before the public launch.

8.2.2 Importance of Prototyping in the Product Development Process

- **Visualization and Communication:** Prototypes provide a tangible way to communicate ideas to stakeholders, team members and potential users. They make abstract ideas concrete and help everyone understand the vision.
- **User-Centered Design:** Prototyping allows designers to gather feedback from end-users early, which ensures the final product addresses real user needs and preferences.
- **Risk Reduction:** Developers can identify and fix design flaws, technical challenges and usability issues before investing significant resources in full-scale production, which reduces the likelihood of costly errors.

- **Iterative Development:** Multiple versions of a product are created and improved based on feedback. This leads to a more refined and user-friendly end product.
- **Cost Savings:** It is cheaper to find and fix problems during prototyping than after the product has been manufactured or deployed.
 - *Example:* Changing a cardboard model costs almost nothing, while changing a factory mould is very expensive.
- **Investor and Stakeholder Confidence:** High-fidelity prototypes give investors a realistic preview of the final product's capabilities and potential, helping to secure funding.

In summary: prototypes are invaluable tools that allow teams to explore ideas, engage with users and refine concepts to create products that are more likely to succeed in the market.

8.3 Creating Prototypes

8.3.1 Tools and Materials for Creating Prototypes

The choice of tools and materials depends on the type of prototype you are building and your project's requirements.

- **1. Sketching and Paper Prototypes:**
 - **Sketchbooks and Paper:** pencils, pens, markers and paper are essential for low-fidelity paper prototyping.
- **2. Digital Prototyping:**
 - **Design Software:** graphic design or prototyping software like Adobe XD, Sketch, Figma or InVision to create interactive digital prototypes with realistic user interfaces.
 - **Coding:** for high-fidelity interactive prototypes, web and app developers may use HTML, CSS and JavaScript to create functional prototypes.
- **3. 3D Modeling and Printing:**
 - **3D Modeling Software:** tools like Autodesk Fusion 360 or Blender are used to create 3D models of physical products.
 - **3D Printers:** transform digital designs into tangible physical objects.
- **4. Materials for Physical Prototypes:**
 - **Clay and Sculpting Materials:** useful for sculptures or models of product designs.
 - **Cardboard and Foamboard:** excellent for building low-cost, low-fidelity physical prototypes.
 - **Wood, Plastic or Metal:** commonly used for more durable, high-fidelity physical prototypes.
- **5. Prototyping Kits:** Specialized kits that include a combination of materials, tools and components, such as Arduino kits for electronics prototypes.
- **6. 3D Scanners:** Devices used to scan existing objects and convert them into digital models for modification or replication.

8.3.2 Building Low-Fidelity Prototypes

Low-fidelity prototypes are essential for quickly visualizing and testing early-stage concepts.

- **1. Sketching and Paper Prototypes:**
 - Start with rough sketches and drawings of your product or interface on paper. Keep it simple and focus on layout and basic functionality.
 - Use sticky notes or index cards for interactive elements and simulate user interactions by manually moving or swapping them.

- Conduct usability tests with team members or potential users to gather feedback and make improvements.
- **2. Digital Prototyping (Low-Fidelity):**
 - Create wireframes or mockups using design software. These are simplified representations of the final product with minimal details.
 - Use placeholder text and generic icons to convey content and functionality.
 - Test the digital prototype with stakeholders or users to get feedback on the overall layout and user flow.
- **3. Physical Prototypes (Low-Fidelity):**
 - Use readily available materials like cardboard, foamboard or clay to build physical mockups.
 - Focus on basic form, size and functionality while keeping aesthetics secondary.
 - Test the physical prototype for ergonomics, usability and fit within the intended environment.

The 8 most common prototyping methods:

- Sketches and Diagrams
- Paper Interfaces
- Storyboards
- Lego Prototypes
- Role Playing
- Physical Models
- Wizard of Oz (faked) Prototyping
- User Driven Prototypes

8.3.3 Iterative Design and Refinement

Prototyping is an iterative process that involves creating, testing and refining your designs multiple times.

- **1. Gather Feedback:** Collect feedback from users, team members and stakeholders after each round of testing. It should focus on usability, functionality and overall user experience.
- **2. Analyze Data:** Analyze the data and observations from usability tests to identify areas of improvement. Look for common pain points, issues and patterns in user behavior.
- **3. Iterate:** Based on the feedback and analysis, make the necessary changes: revise the design, alter the user interface or address technical issues.
- **4. Repeat Testing:** After making revisions, test the updated prototype again. Continue the cycle of testing, analysis and refinement until the prototype meets the desired goals and user needs.
- **5. High-Fidelity Prototyping:** As the design matures and you gain confidence in the concept, move to high-fidelity prototypes with more realistic details such as graphics, colors and interactive elements.
- **6. Final Validation:** Use high-fidelity prototypes for final validation with users and stakeholders. Ensure the product aligns with expectations and is ready for further development stages such as production or coding.
- **7. Documentation:** Document the changes and decisions made during each iteration to keep a record of the design evolution. This is valuable for reference and future development.

Fig 8.4 (Iterative design): Brief/spec → Concept → **First iteration:** paper prototype → **Second iteration:** hard-coded prototype → **Third iteration:** database-driven prototype → final version. A test is carried out after every prototype.

8.4 Testing Prototypes

8.4.1 Importance of User Testing

User testing is a critical phase in the product development process. It serves several essential purposes:

- **User-Centered Design:** Ensures the final product aligns with the needs, preferences and expectations of the target audience, leading to a more satisfying and functional product.
- **Identifying Issues:** Allows you to find and address usability problems, technical glitches or design flaws early, saving time and resources by preventing costly changes later.
- **Validation:** Validates the assumptions and design decisions made during development. It confirms whether the concept effectively solves the identified problem or meets the intended goals.
- **Feedback Collection:** Real users provide valuable insights into their thoughts and experiences, which can inform design improvements.
- **Usability Improvement:** Focuses on how users interact with the product and helps improve the interface, navigation and overall user experience.
- **Iterative Refinement:** Test results drive an iterative design process, allowing incremental improvements until the product meets or exceeds user expectations.

8.4.2 Analyzing and Interpreting Test Results

- **1. Data Collection:** Collect both quantitative data and qualitative data during testing.
 - **Quantitative:** task completion rates, time spent, survey responses.
 - **Qualitative:** user comments and observations.
- **2. Data Analysis:** Organize and analyze the data to identify patterns, trends and outliers. Look for common usability issues or areas where users struggled.
- **3. Prioritization:** Rank the issues by their severity and impact on user experience. Address critical issues first.
- **4. Iterative Changes:** Use the results to make changes to the prototype, such as adjusting the interface, improving navigation or fixing technical bugs.
- **5. Validation:** Re-test the prototype after making changes to make sure the issues are resolved and the user experience has improved.
- **6. Documentation:** Record the results of each testing phase, the changes made and the reasons behind them. This helps track progress.
- **7. Continuous Improvement:** Testing and refinement are ongoing processes. Continue to test and iterate until the prototype reaches the desired level of usability, functionality and user satisfaction.

By conducting user testing and analyzing the results effectively, your prototype evolves into a final product that meets user needs and provides an exceptional user experience, increasing the likelihood of success in the market and fostering customer satisfaction and loyalty.

8.5 Building and Launching the MVP

8.5.1 Developing the MVP (Minimum Viable Product)

A Minimum Viable Product (MVP) is a critical concept in product development. It is the **most basic version** of your product that includes only the essential features needed to address the core problem or need of your target audience.

- **Feature Prioritization:** Identify the core features that are absolutely necessary to address the primary problem. These features should provide value to early users.
 - *Example:* For a food delivery app, the MVP may only have menu browsing, ordering and cash-on-delivery. Loyalty points and live tracking can come later.
- **Simplicity:** Keep the MVP as simple and lean as possible. Avoid feature creep (adding too many extra features) and focus on a streamlined user experience.
- **Rapid Development:** Use agile development methodologies to build the MVP quickly. This allows flexibility in responding to changing requirements and user feedback.
- **Testing:** Continuously test the MVP with real users to gather feedback and validate assumptions. Iterate and make improvements based on user insights.
- **Scalability:** Although the MVP is minimal, design it so it can scale in the future. Make sure the architecture and technology choices support future enhancements.
- **User Onboarding:** Pay attention to onboarding and guidance within the MVP. Make it easy for users to understand and use the product effectively.

Successful Business Examples

- **Facebook:**
 - **Approach:** Started as a social networking platform for college students but followed an iterative development approach. It continuously added features, expanded to different user demographics and acquired companies like Instagram and WhatsApp.
 - **Success:** Now one of the largest social media platforms globally, with over 2 billion monthly active users. Its iterative approach allowed it to adapt to changing user needs and preferences.
- **Amazon:**
 - **Approach:** Began as an online bookstore and gradually expanded into a vast e-commerce platform. It experimented with new features such as user reviews and recommendations and expanded into new product categories.
 - **Success:** One of the world's largest online retailers, offering a wide range of products and services including Amazon Web Services (AWS) and Amazon Prime. Its focus on continuous improvement and customer-centric design is key to its success.
- **Netflix:**
 - **Approach:** Started as a DVD rental service and transitioned into a streaming platform. It continuously analyzed user data to improve its recommendation algorithms and content offerings.
 - **Success:** A global streaming giant with millions of subscribers worldwide. Its iterative approach to content production and personalized recommendations has revolutionized the entertainment industry.

Successes and Failures: Lessons Learned

- **Twitter's Early Failures:**

- **Challenge:** Technical issues and outages in its early years due to rapid user growth; scalability and infrastructure were major obstacles.
- **Lesson Learned:** Perseverance and continuous improvement allowed it to fix these issues. It learned the importance of investing in infrastructure and scalability to support a growing user base.
- **Snapchat's User-Driven Features:**
 - **Challenge:** Criticized for its seemingly simple, ephemeral messaging concept. It needed to keep innovating to stay relevant and compete with larger platforms.
 - **Lesson Learned:** Its iterative approach introduced Stories, Filters and Discover, which suited user preferences. It learned the importance of listening to its young user base and adapting to their evolving needs.
- **Airbnb's Trust and Safety Concerns:**
 - **Challenge:** Trust and safety issues when hosts and guests faced problems such as property damage and security concerns.
 - **Lesson Learned:** It invested in strong trust and safety measures and kept improving the platform based on user feedback, showing the importance of user safety and trust even in the sharing economy.
- **Uber's Global Expansion:**
 - **Challenge:** Regulatory challenges and resistance in various markets during its rapid global expansion.
 - **Lesson Learned:** It learned to adapt its business model and strategies to local regulations and cultural differences, needing flexibility and collaboration to overcome market-specific challenges.

These case studies illustrate the power of iterative development and continuous improvement. They also highlight the importance of **resilience, adaptability and user-centricity** in overcoming challenges and achieving long-term growth.

Pakistan's Successful Business Startups

- **Tajir:** A business-to-business (B2B) platform for small businesses.
- **GrocerApp:** A low-cost online grocery store where customers order products from many categories and get them delivered to their door.
- **Dawaai:** A digital health platform that includes an online pharmacy network and generic pharmaceuticals.
- **Finja:** A financial services platform providing professionals and SMEs with payment, loan and collection services.
- **Salesflo:** A B2B platform that allows retailers to place orders directly with manufacturers, wholesalers and authorized distributors.
- **Daraz.pk:** An online retailer selling branded clothing, footwear, apparel, jewelry and accessories for men and women.
- **Bykea:** A social impact app that saves clients time and money by providing instant transport and courier services.
- **Cheetay:** Pakistan's fastest-growing NOW-Commerce platform with five verticals to choose from.
- **SadaPay:** A "neobank" in Pakistan.

- **TAG:** Pakistan's first digital bank.
- **Zameen.com:** Pakistan-specific software that helps people navigate the real estate market.
- **CarFirst:** A marketplace in Pakistan for selling second-hand vehicles quickly.

Pakistani Entrepreneur Example: Neuro-Mat

Neuro-Mat is a classic example of an **e-health startup**. It is a comfortable, multi-sensory mat equipped with sensors so that basic parameters of an elderly person or a patient can be monitored while resting.

- **What it does:** Provides comfort and also collects data using sensors to monitor the movements of the elderly and patients. It has an alarm (buzzer) that alerts attendants and health staff when the patient leaves the mat.
- **Motivation:** The startup was inspired by the small number of healthcare staff in certain places. They wanted to reduce the tedious task of continuous patient monitoring without staying physically with the patient. Absence of a support staff member can reduce the elderly person's mobility and increase the risk of injury. With Neuro-Mat, healthcare staff can support multiple patients at the same time.
- **Cost breakdown (Fig 8.9):** The electrical circuit (pressure sensors, breadboard, buzzer, wires, micro-controller) is about **50%** of the cost, foam sheets **35%**, and transport and logistics **15%**. So sensors and circuitry are around half, while material and logistics are the other half.
- **Smoke tests (Fig 8.10):** A smoke test checks interest in a product before building it (here with a Google Form and landing page).
 - 45 out of 52 respondents were interested in pre-ordering.
 - 43 believed senior citizens and dementia patients need constant supervision.
 - It was viewed as a cheaper alternative to hiring caretakers.
 - Compared with baby monitors (score 8) and hired caretakers (score 14), Neuro-Mat scored 30, around double the nearest competitor (hired caretakers).
 - Constant monitoring was a booster and, due to its one-time purchase cost, it was treated as a cheap solution. More than 85% of users who tried it in the smoke test showed interest in pre-ordering, a huge success.
- **Feedback and survey (Fig 8.11 and 8.12):**
 - People gave positive feedback, e.g. a cheaper alternative to caretakers and useful for families who cannot give 24/7 care.
 - 36 respondents preferred to **buy** and keep it at home for continuous use or as a backup, rather than **rent** it (16).
 - People were interested in a monthly subscription model, and cash-on-delivery was the preferred mode of payment.
 - Price preference: Rs. 2500-3000 (20 people), Rs. 3000-3500 (17), Rs. 3500-4000 (4), Rs. 4000+ (11).
- **Lean experiment iterations (Fig 8.13):** To make the product multi-variable (serving different audiences in different forms), a series of experiments was done:
 - **1. Final product idea.**
 - **2. Complex circuit:** pressure sensors were distributed throughout the mat, and the computer program that collected and analyzed data became complex.

- **3. Cost-driven circuit:** micro-controllers and infra-red sensors replaced the existing devices and were re-tested for similar and better results.
- **4. MVP:** a small-scale circuit with a buzzer and limit switch for the convenience of support and health staff.
- **Future demand (Fig 8.14):** There are currently over 500,000 diagnosed dementia patients in Pakistan. The projected number of people with dementia (in thousands) is 430 in 2015, 712 in 2030 and 1,422 in 2050. Dementia patients are the primary audience, so demand for Neuro-Mat will rise.

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Short Questions

1. What is a prototype?

Answer: A prototype is a tangible or digital preliminary model of a product, made during the design and development phase, that lets designers, engineers and stakeholders visualize, test and refine the concept before full-scale production.

Explanation: It provides feedback and insights that help improve the final product.

2. Differentiate between a product and a service.

Answer: A product is a physical item that can be seen, touched or held (e.g. smartphone), while a service is intangible and involves an action or task performed by individuals or organizations (e.g. healthcare, consulting).

Explanation: Both are created to meet a customer need or want.

3. Give three reasons why successful product development is important.

Answer: Competitive advantage, revenue generation and customer satisfaction.

Explanation: Other reasons are innovation and adaptation, and market expansion.

4. Name the types of prototypes.

Answer: Low-fidelity (sketches and wireframes), medium-fidelity (paper prototypes), high-fidelity (digital prototypes), functional (proof-of-concept) and working prototypes (alpha and beta).

Explanation: They differ in detail, cost and purpose.

5. What is the difference between alpha and beta prototypes?

Answer: An alpha prototype is an advanced, functional version used for internal testing and refinement. A beta prototype is a near-final version given to a select group of users to find remaining bugs before the official release.

6. Give three benefits of prototyping.

Answer: Risk reduction, cost savings and early user feedback (user-centered design).

Explanation: Fixing a problem on a prototype is much cheaper than after manufacturing.

7. Name four tools or materials used to create prototypes.

Answer: Paper and pens, Figma/Adobe XD, 3D printers, and cardboard or foamboard. Arduino kits and 3D scanners are other examples.

8. What is a low-fidelity prototype and why is it used?

Answer: A simple, quick and inexpensive prototype with minimum visual details, such as a sketch, wireframe, paper or cardboard model.

Explanation: It is used to quickly test early-stage ideas, layout and basic functionality.

9. List the steps of iterative design and refinement.

Answer: Gather feedback, analyze data, iterate, repeat testing, high-fidelity prototyping, final validation and documentation.

10. Why is user testing important?

Answer: It ensures the product matches user needs, identifies issues early, validates design decisions, collects real feedback, improves usability and drives iterative refinement.

11. What is the difference between quantitative and qualitative data in testing?

Answer: Quantitative data is numerical (task completion rates, time spent, survey responses), while qualitative data is descriptive (user comments and observations).

12. What is an MVP?

Answer: A Minimum Viable Product is the most basic version of a product with only the essential features needed to solve the core problem of the target audience.

Explanation: It lets a business test its idea quickly with real users at low cost.

13. What is feature creep?

Answer: Adding too many extra features to a product, which makes it complex and delays delivery.

Explanation: MVP development avoids it by keeping things simple and lean.

14. What is a smoke test? How was it used for Neuro-Mat?

Answer: A smoke test measures customer interest before the product is fully built. For Neuro-Mat, a Google Form and landing page showed that 45 out of 52 respondents were interested in pre-ordering.

15. What lesson did Uber and Twitter teach startups?

Answer: Uber showed the importance of adapting the business model to local regulations and culture. Twitter showed the importance of investing in infrastructure and scalability to support growth.

Extensive Questions

(Remember: the answers give the basic concept and bullet points; write more explanation if needed.)

1. What is prototyping? Explain the different types of prototypes with examples.

- **Definition:** A prototype is a tangible or digital preliminary model of a product, used to visualize, test and refine it before full-scale production.
- **Types:**
 - **Low-fidelity:** sketches and wireframes; quick, cheap, minimum detail. Example: hand-drawn app screens.
 - **Medium-fidelity:** paper prototypes; physical interactive models for user testing. Example: paper cut-outs of screens.
 - **High-fidelity:** digital, interactive simulations close to the final product. Example: a clickable Figma prototype.
 - **Functional:** proof-of-concept prototypes that test technical feasibility. Example: a basic sensor circuit.
 - **Working:** alpha (internal testing) and beta (select users before release) prototypes.

2. Discuss the importance of prototyping in the product development process.

- **Visualization and communication:** makes abstract ideas concrete for everyone involved.
- **User-centered design:** early feedback ensures the product meets real needs.
- **Risk reduction:** flaws are found before heavy investment in production.
- **Iterative development:** versions are improved again and again with feedback.
- **Cost savings:** cheaper to fix problems early than after manufacturing.
- **Investor confidence:** high-fidelity prototypes give a realistic preview and help secure funding.

3. Explain the process of building low-fidelity prototypes and the iterative design cycle.

- **Low-fidelity building:**
 - Paper: rough sketches, sticky notes/index cards to simulate interaction, usability tests.
 - Digital: wireframes with placeholder text and generic icons, tested with stakeholders.
 - Physical: cardboard, foamboard or clay; focus on form, size and function; test ergonomics and fit.
- **Iterative cycle:**
 - Gather feedback → analyze data → iterate → repeat testing.
 - Move to high-fidelity prototyping as confidence grows.
 - Final validation with users and stakeholders.
 - Document changes and decisions of each iteration.

4. Why is user testing important? Explain how test results are analyzed and interpreted.

- **Importance:**
 - Ensures user-centered design and finds issues early.
 - Validates assumptions, collects feedback and improves usability.
 - Drives iterative refinement.

- **Analysis steps:**
 - Collect quantitative and qualitative data.
 - Analyze for patterns, trends and outliers.
 - Prioritize issues by severity and impact.
 - Make iterative changes, re-test to validate.
 - Document everything and keep improving continuously.

5. What is an MVP? Describe the key steps in developing it.

- **MVP:** the most basic version of a product with only the essential features that solve the core problem.
- **Key steps:**
 - Feature prioritization: pick only the absolutely necessary features.
 - Simplicity: stay lean and avoid feature creep.
 - Rapid development: use agile methods to build quickly and stay flexible.
 - Testing: test continuously with real users and iterate.
 - Scalability: design so future enhancements are possible.
 - User onboarding: make the product easy to understand and use.

6. Explain the Neuro-Mat startup as an example of prototyping, testing and MVP development.

- **Idea:** a sensor mat that monitors elderly people and patients and sounds a buzzer when they leave the mat, helping staff who are few in number.
- **Feasibility:** circuit about 50% of cost, foam 35%, logistics 15%.
- **Testing:** smoke test showed 45 of 52 interested in pre-ordering; scored about double the nearest competitor (30 vs 14 for hired caretakers).
- **Survey:** most people wanted to buy (36) rather than rent (16), liked monthly subscription and cash-on-delivery.
- **Iterations:** complex pressure-sensor circuit → cost-driven infra-red and micro-controller circuit → MVP with buzzer and limit switch.
- **Demand:** over 500,000 dementia patients in Pakistan, projected to reach about 1.42 million by 2050.

7. Describe how Facebook, Amazon and Netflix used iterative development to succeed. What lessons do Twitter, Snapchat, Airbnb and Uber teach?

- **Facebook:** college network → new features, new demographics, acquisitions (Instagram, WhatsApp); over 2 billion monthly users.
- **Amazon:** bookstore → reviews, recommendations, new categories, AWS and Prime.
- **Netflix:** DVD rental → streaming; used user data to improve recommendations.
- **Twitter:** invest in infrastructure and scalability.
- **Snapchat:** keep innovating (Stories, Filters, Discover) and listen to users.
- **Airbnb:** build strong trust and safety measures.
- **Uber:** adapt to local regulations and culture.
- **Overall:** iteration, resilience, adaptability and user-centricity lead to long-term growth.